# Towards a Model of Volume Transmission in Biological and Artificial Neural Networks: A CAST Approach

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**Abstract.** At present, a new type of process for signalling between cells seems to be emerging, the *diffusion or volume transmission*. The volume transmission is performed by means of a gas diffusion process, which is obtained with a diffusive type of signal (NO). We present in this paper a CAST approach, in order to develop a NO diffusion model, away from a biologically plausible morphology, that provides a formal framework for the establishing of neural signalling capacity of NO in biological and artificial neural environments.
It is also presented a study which shows implications of volume transmission in the emergence of complex structures and self-organisation processes in both biological and artificial neural netwoks.
Finally, we present the diffusion version of the Associative Network (AN) [6], *the Diffusion Associative Network (DAN)*, where a more general framework of neural learning, which is based in synaptic and volume transmission, is considered.

## 1 Introduction

Cellular communication is responsible for neural information transmission and processing, as well as learning and cognitive and behavioural processes in the brain. This rapid and precise communication is made possible by two signalling mechanisms: Axonal conduction and synaptic transmission. Nowadays, the latter is the basic mechanism for excellence, for signalling between cells and for many of the brain processes.
In the synaptic transmission process it can distinguish between basic elements and constraints, which are mutually related. The basic elements are the neurotransmitters, the receptors and the contacts between nerve cells. The constraints

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will be the unidirectional transmission, from pre-synaptic to post-synaptic neuron, the release of neurotransmitters and receptors fit for these neurotransmitters in the pre-synaptic and post-synaptic sides respectively.

At present, a new type of process for signalling between cells seems to be emerging, that is to say, *diffusion or volume transmission*. Volume transmission is performed by means of a gas diffusion process, which is obtained with a diffusive type of signal. The main differences between synaptic and volume transmission is that the latter is membrane permeable and multidirectional because it disregards the spatial constraints on neurotransmitter activity [2].

The presence of a molecule in the brain that acts as a diffusive signal that affects the diverse neighbouring cellular processes opens up new perspectives regarding the study of brain function. Nitric oxide (NO) was soon found to be a more likely candidate, because of its extreme isotropical diffusibility in both aqueous and lipid environments, which allows a rapid three-dimensional spread of the signal irrespective of the presence of membranes [11]

In the absence of relevant experimental data for understanding how NO functions as a neuronal signalling molecule, we have developed a NO diffusion computational model. Several NO diffusion models have been published [4], [5] and [8]. They consider the spread of NO from sources whose morphology are not biologically validated and even some mathematical anomalies and biologically implausible results can be introduced for these approaches.

We present in this paper a CAST approach in order to develop an NO diffusion model, away from a biologically plausible morphology, that provides a formal framework to establish the neural signalling capacity of NO in biological and artificial neural environments. We will address this problem using problem-solving by models [1]. In addition to this, our model also has repercussions on establishing hypothesis regarding the existence of neural mechanisms and the design and interpretation of biological experiments on NO behaviour and effects on brain structure and function.

Furthermore, our study is different as and for first time, we show the emergence of complex structures and self-organisation processes in both biological and artificial neural netwoks.

Finally, we present the diffusion version of the Associative Network (AN) [6], *the Diffusion Associative Network (DAN)*, where a more general framework of neural learning, which is based in synaptic and volume transmission, is considered.

## 2 NO Diffusion Computational Model: A CAST Approach

The main objective of this study is to move towards a better understanding of the underlying neural mechanisms of cellular communication in the brain, their relationship with learning and structure emergence in biological and artificial neural networks.

Our research and this section are focused on a new kind of neural signalling transmission, volume transmission, whose underlying mechanism is a diffusive signalling molecule (NO) working as a retrograde cellular messenger. We will be specifically taking the first step towards developing the Theoretical Framework (TF) belonging to the NO diffusion messenger global study framework (GSFNO) [2].

We present a CAST approach in order to develop the NO diffusion computational model, which provides new neural signalling transmission models. We will address this problem using problem-solving by models [1]. Taking into account the structure of a problem-solving process, we can say that our research belongs to the modelling process stage.

Using the different types of models proposed by Pichler [1], our initial $M_0$ model is of the generator type. By following the model-transformation application method, we obtain a dynamics $M_f$ type model which will be used to model the NO diffusion.

The diffusion process is closely linked to the transport phenomena. This phenomena takes place when, due to the gradient of a physical magnitude, another is displaced and invades and affects its setting in a structural and/or functional way. Its behaviour does not depend on any type of molecular theory and can be studied from a phenomenological point of view. Thus, we propose to use physical phenomenological laws of diffusion, Fick's laws regarding isotropic mediums, Eq. (1) [12], and theoretical and experimental biological studies into the NO generation pattern [2], [4], [9] and [10].

$$F = -D\frac{\partial C}{\partial x} \ . \tag{1}$$

where $F$ is the rate of transfer per unit area of section, $C$ the concentration of diffusing substance, $x$ the space co-ordinate measured normal to the section, and $D$ is the diffusion coefficient.

The fundamental differential equation of diffusion in an isotropic medium is derived from (1)

$$\frac{\partial C}{\partial t} = D\Big(\frac{\partial^2 C}{\partial x^2} + \frac{\partial^2 C}{\partial y^2} + \frac{\partial^2 C}{\partial z^2}\Big) \ . \tag{2}$$

where $x$ ,$y$ y $z$ are the space coordinates.

The dynamics of the diffusing messenger have not yet been experimentally defined. Taking into account the fact that NO release does not need presynaptic specialisation, the whole surface of an NO neuron is a potential source. This implies that these dynamice respond to a volumetric morphology. We propose a cylindrical morphology, which considers the *neuron* as the basic NO module emitter. This obeys the quantal and non-local nature of the NO emission, the possibility of NO emission in multiple areas of the neuron, which leads us to a parallel emitting source structure situated throughout it, as well as to the NO self-regulatory capacity.

We can therefore consider a long circular cylinder throughout which diffusion is radial. Thus, the $M_0$ is given by the Eq. (2) in terms of the cilindrical coordinates, modified by the kinetics of NO decay. They are expressed by means of an inactivation function, used to model the NO loss through various chemical reactions. This has been taken to be exponential decay since there is not real data to suggest any other function. Thus: $1/2life = ln(2/\gamma)$ [8].

$$\frac{\partial C}{\partial t} = \frac{1}{r}\frac{\partial}{\partial r}\Big(rD\frac{\partial C}{\partial r}\Big) - \gamma C \ . \tag{3}$$

Using the model-transformation application method, we obtain a *dynamics* type model $M_f$ (4), in terms of Bessel functions [7], suitably chosen so that the initial and boundary conditions are satisfied.

$$C(r,t) = \sum_{n=1}^{\infty} A_n J_0(\xi_n r) e^{-\lambda_n Dt} \ . \tag{4}$$

where

$$A_n = \frac{2}{r_1^2 J_1^2(r_1\xi_n)} \int_0^{r_1} f(r) J_0(\xi_n r) r \mathrm{d}r \ . \tag{5}$$

If the initial concentration distribution $f(r)$ is maintained at zero concentration and the surface $r = r_1$, then $J_0(x)$ and $J_1(x)$ are Bessel functions of the first kind of order zero and one, $\lambda_n - (\gamma/D) = \xi_n^2$. Where $\lambda_n$ is the separation variable, and the diffusion constant is $D = 3,3x10^{-5}cm^2/s$. [11]
Another phenomenological approach, which is not only mathematically valid, but also biologically coherent, used in our model has been to consider the NO synthesis process as a pulse function. This coherence obeys to experimental results which infer that NO once sinthetised, it freely diffuses [13].

### 2.1 Implications in Neural Networks

The proposed volumetric transmission model gives results that are in line with experimental results, as well as with hypothesis about NO diffusion behaviour, and its possible implications neural architecture and learning processing both in biological and artificial neural networks.

**Fast Diffuse Neural Propagation (FDNP)** The Fig. 1 show NO concentration profiles at various times, t = 0.1s and t=0.2s, according to the distance to the NO neuron and temporal concentration profiles at various distances, r=100$\mu$ms, r=150$\mu$ms. The local and highly dynamic nature of NO distribution can be seen there, which is limited by the diffusion radius that is defined by the half-life of the diffused NO. Therefore, we can observe the existence of a FDNP. In Fig. 2, it can likewise be seen that the presence of diffused NO is slightly stronger in t=0.2s than in t=0.1s in the areas located over 70 micrometres.

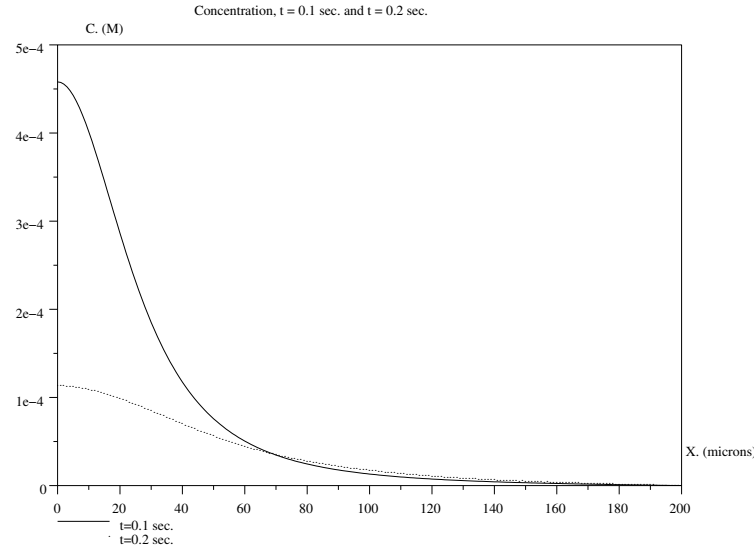



**Fig. 1.** Profiles of the concentration of NO at various times, t = 0.1 $s$., t = 0.2 $s$.

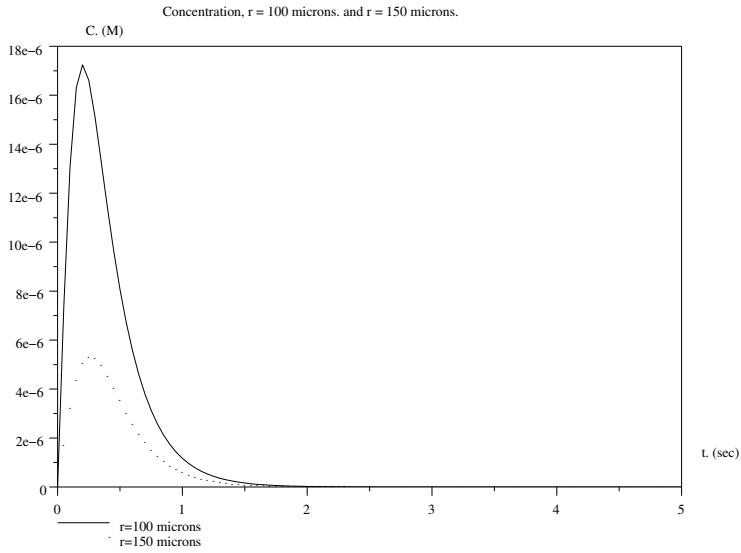



**Fig. 2.** Temporal concentration profiles at various distances, r = 100 $\mu m$., r = 150 $\mu m$.

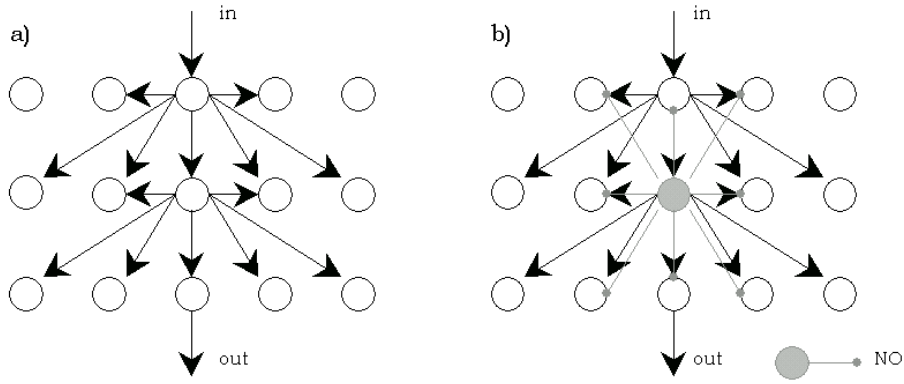



**Fig. 3.** a.- Classical neighbourhood 2-D with synaptic connections. b.- First rate DNB 2-D. (Adapted from Regidor J. and Suarez Araujo C.P., Brain Processes, Theories and Models, Cambridge, Massachusetts, Londo, MIT Pres, 1995

**Diffuse NeighBourhood.** A further result that confirms the reliability of the proposed model is its behaviour in relation to the formation of neural neighbourhoods and the emergence of complex structures. One of the most important NO implications for ANNs is the emergence of Diffuse Neighbourhood (DNB) produced by a diffusion neural association. In this DNB, there will be transmission of some information from the NO neuron to all the neighbouring cells without any synaptic connections being established among them. We can define two kinds of DNB that are directly influenced by the NO neuron: The first-rate DNB, defined on the same neural level, (see fig. 3), and the second-rate DNB on the adjacent anterior and posterior neural levels [2].

The emergence of DNB and three-dimensional spread of the retrograde messenger effect can produce *volume plasticity*. The consequence of this volume plasticity is that correlation will be reinforced between elements which are coactive within a given time window and are within a critical volume, without being necessarily physically connected. This is a step towards obtaining simple neural structures and towards speeding up the learning and network operation. The effect demonstrated by this behaviour may be able to produce self-organisation in the neural network and to build neural columns, in the same way as in the cortex, which has a columnar organisation [2].

In this paper, we describe the existence of first-rate DNB generated by NO diffusion, according to the proposed model. Fig. (4) shows the DNB corresponding to Fig. (5)b, for the times of the NO concentration profiles in Fig. 1, as well as for the parameter experimental values, half-life time, $\gamma$, and maximum diffusion radius, $r_1$ de 200 $\mu m$. and 5 s respectively. These are the parameters that define, for excellence, the form and scope of the diffuse neighbourhood (DNB).

The NO synthesis and, therefore, its diffusion through the neural tissue, are not a static or isolated process and are in fact a dynamic and multiple one. On

the other hand, taking into account the dense framework of the biological neural network and, therefore, the one that the artificial neural network wishes to maintain, the number of neurons that are emitting NO in narrow time bands can be very high, and therefore multiple numbers of DNB are produced. This involves complex structure formation, due to the formation of complex diffuse neighbourhoods, emerging both from the high time and space neighbourhood overlapping. This phenomena is demonstrated by our model, considering the lineal superposition between the various emerging DNB. Figs. (5)a and (5)b show the complex structure formation when 2 and 3 DNBs emerge as a result of various diffusion process with a 0.1s time interval between them.

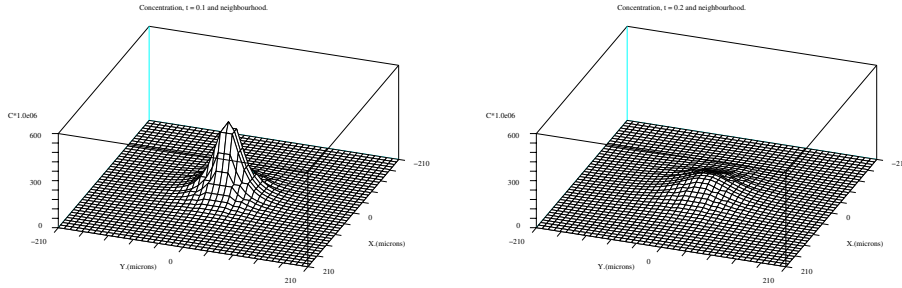
In these neural settings, where a great number of diffuse neighbourhoods are involved, it is very useful to establish the NO neuron direction and therefore the neural assamblies which influence a neuron. That is to say, it allows us to determine which are the diffuse neighbourhoods that influence the behaviour of a specific neuron. This is highly important in the retrograde character of the neuro-messenger. Our model has repercussions for establishing hypothesis on the existence of underlying neural mechanisms to detect NO neuron direction. It has this potential and provides a mathematical formalism that allows us to study the direction of the NO neuron. This formalism is depicted in the so-called "NO neuron detector functions" expressed in (6).

$$\begin{array}{l} f_1(x,y) = |C(x,y) - C(x+dx,y)| \\ f_2(x,y) = |C(x,y) - C(x+dx,y+dy)| \\ f_3(x,y) = |C(x,y) - C(x+dx,y+dy)| \\ f_4(x,y) = |C(x,y) - C(x-dx,y+dy)| \end{array} . \tag{6}$$

The meaning of these functions is determined by the meaning of an increase being produced in them. Taking into account the equations given in (6) and Fig. (6), an increase may be inferred in some of the previous $f_1$ functions, for example, it establishes that the concentration of diffused substance, in (x, y) y (x+dx, y) is changing. This change in the concentration will be caused by diffusion phenomena located in the direction of $f_1$, which implies that the neuron(s) responsible for the generated volumetric transmission, are localised in the direction indicated by this functions. We can apply similar arguments to the rest of the directions determined by functions $f_2$, $f_3$ y $f_4$. Therefore and by means of the behaviour of the detector functions throughout time, a localisation of the diffusion phenomena can be performed.

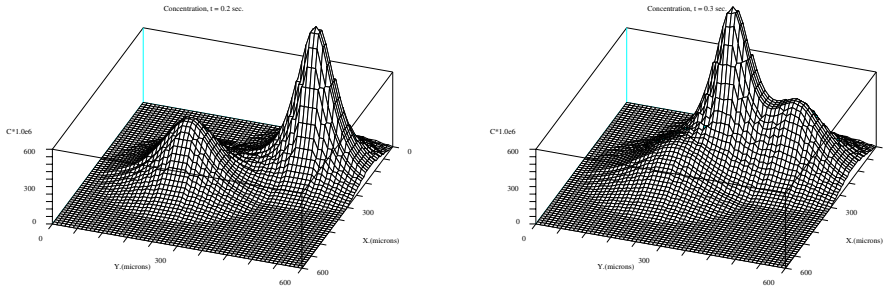
**Learning Processing.** The most appealing characteristic of ANNs is their learning capacity. In this way the ANNs behaviour is emergent of structural changes directed by learning laws which provide plasticity to these neural structures. A natural substratum for these characteristics seems to be present in the diffusive behaviour of NO and in their implications in neural structure and processing.

The learning mechanism both in biological and artificial neural networks has been explained by means of synaptic plasticity using the concept of neurotrans-




(a) DNB at t = 0,1 $s$. (b) DNB at t = 0,2 $s$.

**Fig. 4.** Diffusion Neighbourhoods for a NO neuron at position (0,0).




(a) (b)

**Fig. 5.** Emergence of complex structure for overlapping of several DNBs corresponding to several NO neurons involved in diffusion processes with a delay of 0.1 s. between them a) overlapping of DNBs produced by two NO neurons. b) overlapping of DNBS produced by trhee NO neurons.

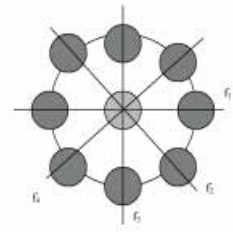


**Fig. 6.** Detectors of direction of NO neurons. NO receptor neuron in the central postion surrounded by 8 NO neurons.

mitters and weights respectively. In both cases it is necessary to establish exact connections between the neurons which are participating in the process. Some neural activities resulting in plasticity, learning or memory can be more easily explained if a retrograde cellular messenger, such as NO, can be considered. Interestingly, NO has been implicated in long-term potentiation (LTP) in the hippocampus [2] and [10]. Nitric oxide diffuses over large distances in cortical tissue. Because of this it may underlie a form of non-local, non-synapse-specific learning. This non-locality leads to issues that are commonly ignored in Hebbian learning rules [2].

The first explicit statement of the physiological learning rule for synaptic modification was given by Donald Hebb in 1949 [3] who tried to explain learning and memory processes. Hebb's assumption, known as the *Hebb synapse*, at the moment called *Hebb learning law*, can be stated as follows: When an axon of cell A is near enough to excite a cell B and repeatedly or persistently takes part in firing it, some growth process or metabolic change takes place in one or both cells such that A's efficiency, as one of the cells firing B, is increased [3].

There is not direct evidence of the cause of the postulated change of the efficiency between two connected cells, A and B. We propose as a probable biological support to the Hebb learning law the presence of an NO retrograde cellular messenger. This neuromessenger is produced during the depolarization process of postsynaptic neuron. It freely diffuses through membranes affecting all neighbouring cells, including the presynaptic one and the cell that released it. Because of this we must take into account a new conception of the meaning of the Hebb law; it is not a correlational postulate any more.

This new conception of Hebb's rule implies a reformulation of its mathematical expression. The change in NO concentration has to be considered as a new variable with an important role in the learning process, and the incursion relation as the formal framework for expressing this learning process.

$$w_{AB}^{t+1} = w_{AB}^{t} + \delta\Big(g_1(x_A(t), \Delta n_A) g_2(S_B(t), \Delta n_B, w_{AB}^{t+1})\Big) \ . \tag{7}$$

where $t$ is discrete time, $w_{AB}$ is the interconnection weights between A and B nuerons, $x_A$ is the information from the presynaptic neuron, $s_B$ is the activation state of a postsynaptic neuron, $\delta$ is the learning rate which is related to the gas diffusion process, $\Delta n_A$ and $\Delta n_B$ are the gradients of NO concentration around neurons A and B respectively. These gradients will be determined by means of transport phenomena using phenomenological laws, as diffusion equations. Finally, $g_1$ and $g_2$ can be linear or non-linear functions which indicate the specific effect of pre and postsynaptic information in the learning process.

We present the effect produced in a learning process due to volume transmission. We propose the simplest modification of Hebb's learning law in a neural architecture where the modification of weights is based on the correlation.

## 3 Diffusion Associative Network (DAN)

In this section, we describe the behaviour of an artificial neural network, both in the learning mode and functioning mode, when the existence of the synaptic and volumetric transmission is considered. We will develop the diffusion version of a classic architecture, the Associative Network (AN) [6], using the model proposed for volumetric transmission. The learning law of the proposed new neural architecture, known as Diffusion Associative Network (DAN), will be of the form given in the equation (9) and it will specifically reflect its most simple abstraction. The Associative Network is based on correlation matrix model [6]. It has an input layer and an associators layer. The input layer, consists of two parts: the key field $\{a_1,a_2,a_3,\ldots,a_n\}$ used for the encoding of data and the data field $\{u_1,u_2,u_3,\ldots,u_m\}$. All input elements are called receptors. The associators consists of a set of elements labelled by a pair (i,j) corresponding to the ith element of the key field and the jth of the data field to which the particular associator is connected.
The learning process to memorise data consists of increasing the value of every associator element $M_{ji}$ by the amount directly proportional to $a_i u_j$.
For a set of patterns $(\mathbf{a}^1,\mathbf{u}^1),(\mathbf{a}^2,\mathbf{u}^2),(\mathbf{a}^3,\mathbf{u}^3),\ldots,(\mathbf{a}^p,\mathbf{u}^p)$,

$$M_{ji} = \rho \sum_p a_i^p u_j^p \ . \tag{8}$$

where $\rho$ is the normalising constant.
The recall of a particular $u_i^r$ associated with a key vector $\mathbf{a}^r$ is made by transformation

$$\hat{u}_j^r = \sum_i M_{ji} a_i^r \ . \tag{9}$$

The structure of the Diffusion Associative Network (DAN) is shown in the Fig. (7). The associator elements are organised in neural assemblies. The number of neural assemblies in a DAN is equal to the dimension of the key field, n. Each assembly is made up by a number of associator elements equal to the size of the data field of the DNA. Each associator belonging to an ith neural assembly receives two connections from the receptor field. One comes from the ith receptor of the key field, which is the one that defines the neural assembly, and the other connection is received from the receptor belonging to the data field, of the same order as the associator. An associator element of the DNA is defined by $M_{ij}$, where the subscript $i$ indicates the ith neural assembly, and the subscript $j$ the jth associator element of the assembly. Thus, a jth associator element of the ith assembly receives the input of the $u_j$ and $a_i$ elements. Fig. 7.
This artificial neural network involves the influence of neural structures in each of its neurons, not only due to the synaptic transmission by means of the specified synaptic connections, but also to the volumetric transmission by means of the diffuse neighbourhoods (DNB). In this way, an assembly will exercise an influence on the other assemblies that are in the DBN. Therefore, in this monodimensional

structure of neural assemblies, we have selected influences such as the ith assembly, which can be influenced by the neural assemblies (i-1)th, ith and (i+1)th, during the diffusion period. This involves the existence of neural information transmission between assemblies supported by the volumetric transmission and therefore volumetric plasticity and a *volume learning* [2].
A generalisation of this model would involving considering that DNB intra assemblies and not only inter-assemblies exist. Our model includes the inter-assembly volumetric transmission. This is expressed in equations (10) and (11), where the DNA learning process is formalised, which allows the data submitted to it to be learnt, which is a set of $p$ data in this case. This equation can be seen as a special particularization, discreet in time, of the equation (7). Where $\delta = \rho$ are represented by the parameter $c$ obtained from our model in the discrete of the volumetric transmission, and $g_1$ and $g_2$ are chosen in such a way that represents the correlation between the input and output.

$$M_{ji} = \rho \sum_{p} c_i^p a_i^p u_j^p \ . \tag{10}$$

In this case the correlation is weighed by $c_i^p$ which has a behaviour based in the fiffusion [5]

$$c_i^p = (1 - 2D) f(a_i^p) + D(c_{i-1}^{p-1} - c_{i+1}^{p-1}) \ . \tag{11}$$

and where

$$f(a_i^p) = \begin{cases} \eta & \text{if } a_i^p = max(a_i^p) \\ c_i^{p-1} & \text{otherwise} \end{cases} \tag{12}$$

where $\eta$ is the strenght of the NO neurons.

### 3.1 Results

In this section, we will show that consideration of the existence of fast diffuse neural transmission and therefore volumetric transmission has implications for the learning process. Not only do the results submitted corroborate the reliability of the proposed model, but also that the synaptic and volumetric transmission are the signal transmission mechanisms responsible for learning. On the other hand, it backs the modification of Hebb's law set out in equation (9), as well as the new formal framework for learning in ANN proposed by Surez Araujo [2] and [12].
We submitted the set of patterns shown in the Fig. (8), to both the AN and the DAN networks, so that they were memorised by them. We performed a data recovery process and the results obtained show the great capacity of the DNA compared to the AN for learning and therefore for recovering the data with greater precision. Furthermore, the reduction of the crosstalk phenomena in the reaction of the Diffusion Associative Network is shown. The Fig. (9) shows the

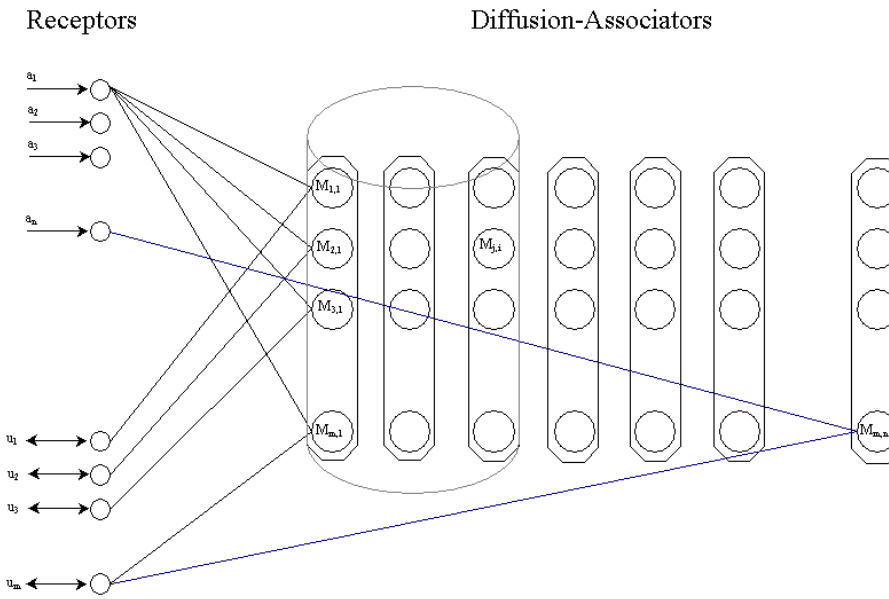


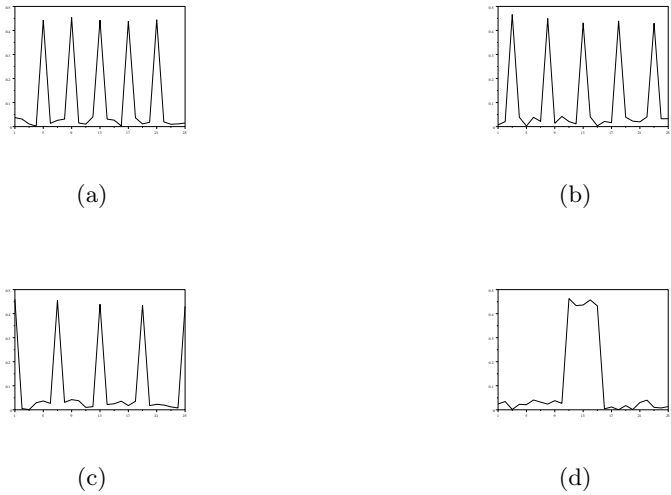

**Fig. 7.** Structure of a Difussion Associative network (DAN).

results of the two associative networks, AN and DAN, for a data recovery process for the patterns shown in Figs. (8)(a) and (8)(d).

## 4 Conclusions

In this paper we present a CAST approach in order to develop biologically plausible model of NO diffusion, moving towards a volume transmission model in biological and artificial scenarios. We have used problem-solving by models and have established a dynamic type model.

This paper, using this mathematic-computational model, has explored the neurobiological and artificial implications of NO as a retrograde cellular messenger, proposed in the first paper belonging to NO and ANNs Series [2]. The dynamics of NO diffusion behaviour has been considered and we have shown the existence of a Fast Diffusion Neural Propagation (FDNP). We have also looked at the emergence of complex neural structures and self-organisation processes by the emerging the Diffuse Neighbourhood (DNB) and multiple temporal/spatial DNB overlapping. This paper has also studied the implications of NO effects on learning processing in general, and on the learning laws in particular. Specific consideration has been given regarding the reformulation of the mathematical expression of Hebbs learning law and another one concerning a new formal framework for learning in ANNs. This framework will consist of two kinds of learning:



(a) (b)

(c) (d)

**Fig. 8.** Set of patterns used in both the AN and the DAN networks.

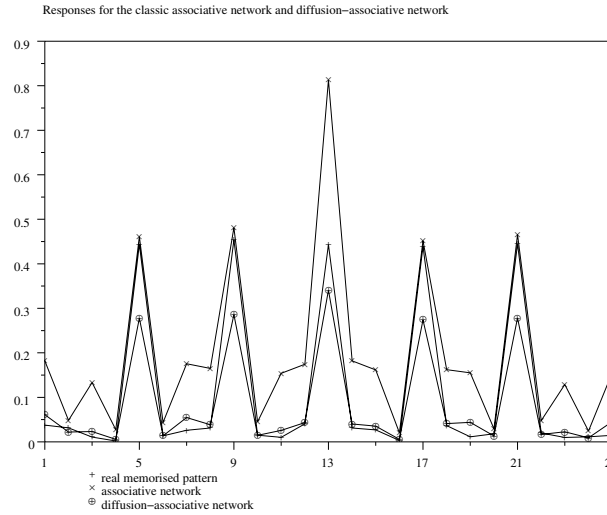
one based on the classical neurotransmission and a further one based on volume transmission.

All these considerations have been taken in an Associative Network (AN). The result has been a new neural architecture, the Diffusion Associative Network (DAN). Thus, we have demonstrated how adding volume transmission mechanisms to an artificial neural network leads to better solutions for real problems. The research in this paper also has repercussions for establishing hypothesis on the existence of underlying neural mechanisms, such as the capacity to detect NO-neuron direction. It also contribute to design and re-interpretation of new biological experiments on behaviour and effects of NO on the brain structure and function.

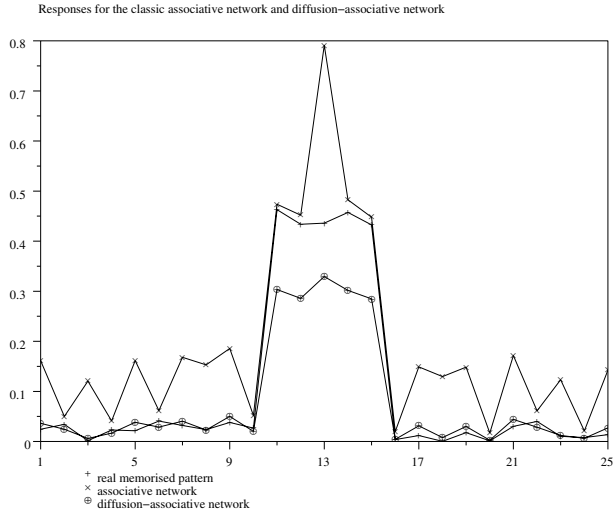
Finally, it can be said that the proposed model and the new artificial neural network, DAN, confirm the important fact that biological structures can supply new ideas in order to obtain more realistic and powerful artificial intelligent systems. In the same way, it can also be said that the computational models can supply new biological hypothesis which address the design and re-interpretation of new biological experiments.

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(a) + real memorised pattern. × associative network. ⊕ diffusion-associative metwork.




(b) + real memorised pattern. × associative network. ⊕ diffusion-associative metwork.

**Fig. 9.** Results of the two associative networks, AN and DAN for a data recovery process for the patterns shown in: (a) Fig. 8(a). (b)Fig. 8(d).

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